

MANOJ PENTAPATI

Software Engineer

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Summary

Computer Science undergraduate specializing in Data Science with strong foundations in Java, Python, C++, and SQL. Skilled in data structures, object-oriented programming, and building complete software applications from design to deployment. Solved **200+ LeetCode problems** and ranked in the **top 1% nationally** in the Naukri Campus Young Turks 2025 assessment. Passionate about clean code, collaborative problem-solving, and delivering software that creates measurable impact.

Technical Skills

Programming Languages: C++, Java, Python, HTML, CSS

Core Concepts: Data Structures and Algorithms, Object-Oriented Programming, Design Patterns

Databases: MySQL (Joins, Aggregations, Subqueries, Indexing), MongoDB

Developer Tools: Git, GitHub, Docker, Visual Studio Code

Experience

Teaching Assistant – Object-Oriented Programming (Java)

January 2026 – Present

Vignan's Foundation for Science, Technology and Research

Vadlamudi, India

- **Delivered lab sessions and supported lectures** on core OOP concepts in Java for **60+ undergraduate students** across 3 lab batches, directly contributing to improved assignment completion rates.
- Guided **15+ students per session** through hands-on debugging and coding exercises on Inheritance, Polymorphism, Abstraction, and Encapsulation, strengthening individual student confidence.
- Collaborated with 2 faculty members to design lab exercises and grading rubrics, **standardizing practical assessments** and improving evaluation consistency across all OOP lab sections.

Projects

Object Tracking System with GUI | *Python, YOLOv5, SORT, OpenCV, Tkinter* | 🐙 [GitHub](#) Jan 2025 – Mar 2025

- **Built a real time object tracking system** using YOLOv5 for detection and the SORT algorithm for persistent tracking, supporting **5+ simultaneous objects** per video frame with stable unique IDs.
- Designed and implemented a Tkinter-based GUI with **4 configurable parameters**, enabling users to upload videos and visualize tracked objects with bounding boxes in real time.
- Optimized the video processing pipeline using OpenCV, **reducing frame processing time by 30%** through efficient frame extraction and object path caching.

Pizza Sales Analysis Using MySQL | *MySQL, SQL, CSV Data Processing* | 🐙 [GitHub](#)

Jul 2024 – Aug 2024

- **Analyzed 21,350 orders and 48,620 order detail records** across 12 months to uncover revenue trends, peak sales hours, and top customer preferences, translating raw data into actionable business insights.
- Designed a normalized relational schema with **4 linked tables**, enforcing foreign key constraints and indexing strategies that **reduced query execution time by 40%** over flat file analysis.
- Applied advanced SQL techniques including window functions, CTEs, and complex joins to identify top-selling product categories and deliver clear, stakeholder-ready sales reports.

Achievements & Certifications

- **LeetCode 200-Day Badge (2025):** Solved **200+ problems** across arrays, graphs, dynamic programming, and trees while maintaining a 200-day consecutive streak.
- **99.49 Percentile – Naukri Campus Young Turks 2025:** Ranked in the **top 1% nationally** among thousands of engineering students in a technical assessment covering DSA, OOP, and databases.
- **OOP Certificate – Take U Forward (TUF), December 2025:** Completed a structured Java OOP course covering design patterns, SOLID principles, and real-world application development.

Education

B.Tech in Computer Science and Engineering (Data Science)

Expected Graduation: May 2026

Vignan's Foundation for Science, Technology and Research, Vadlamudi

CGPA: 8.12 / 10